Interpreting Lenacapavir Resistance from Public Phenotypic Data: Mutation Main Effects Dominate with Negligible Genetic Backbone Modulation

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April 21, 2026

Abstract

Lenacapavir, the first HIV-1 capsid inhibitor, has shown resistance mutations at the NTD-CTD interface (N57H, Q67H, N74D) and hydrophobic pocket (M66I) in clinical trials. We integrated 87 phenotypic measurements from 4 studies (158 mutations, 15 HIV-1 subtypes) and applied hierarchical mixed-effects modeling to decompose resistance into mutation fixed effects and subtype random effects. Results show mutation main effects dominate (N57H: 2951-fold, M66I: 1950-fold) with near-zero between-subtype variance ($\sigma^2 = 1.4 \times 10^{-10}$), indicating resistance is determined by mutations themselves rather than genetic backbone. Three rigorous cross-validation schemes (leave-one-study-out RMSE=1.69, leave-one-backbone-out RMSE=1.66, leave-one-mutation-combination-out RMSE=1.99) demonstrate model extrapolation to unseen data. Sensitivity analysis confirms negligible between-study effects ($\sigma^2 = 6.2 \times 10^{-7}$) and consistency between in vitro and clinical data. Biological anchoring maps high-impact mutations to capsid binding sites, fitness costs (single mutants $< 10\%$ wild-type), and clinical evolution pathways from CAPELLA trials (M66I→M66I+N74D→M66I+N74D+A105T). This framework provides interpretable quantitative analysis of lenacapavir resistance, supporting subtype-independent surveillance strategies.

1 Introduction

Lenacapavir represents the first long-acting HIV-1 capsid inhibitor, offering twice-yearly dosing that transforms treatment adherence. Resistance mutations have emerged at the capsid NTD-CTD interface (N57H, Q67H, N74D) and hydrophobic pocket (M66I) in clinical trials and in vitro selection. Characterizing resistance requires integrating heterogeneous evidence: clinical failures from CAPELLA trials, in vitro dose-response assays, and natural polymorphisms across HIV-1 subtypes A1, B, C, and D.

Prior analyses treated each mutation-subtype combination independently, precluding generalization across genetic contexts. We hypothesized that resistance decomposes into mutation main effects (capturing direct disruption of drug binding) and genetic backbone effects (reflecting subtype-specific capsid stability). Hierarchical mixed-effects modeling enables this decomposition while accounting for study heterogeneity.

2 Methods

2.1 Data Curation

We unified 273 records from 4 sources: PMC12077089 (in vitro), JAC2025 (natural polymorphisms), JID2025 (clinical), and NATAP2022 (mixed). After filtering for quantitative fold-change (FC) and subtype annotation, 87 measurements spanning 158 mutations and 15 subtypes remained. Quality scores (1-3) weighted observations by experimental rigor.

2.2 Hierarchical Model

We fit a linear mixed-effects model:

$$\log_{10}(\mathrm{FC}_{ij}) = \beta_0 + \beta_{\mathrm{mut}_i} + u_{\mathrm{backbone}_j} + \epsilon_{ij} \tag{1}$$

where β_{mut_i} are fixed mutation effects, $u_{\mathrm{backbone}_j} \sim N(0, \sigma^2_{\mathrm{backbone}})$ are random subtype effects, and ϵ_{ij} is residual error. REML estimation via statsmodels MixedLM.

2.3 Validation Strategy

Beyond standard LOOCV, we implemented:

- **Leave-one-study-out**: Tests generalization across data sources
- **Leave-one-backbone-out**: Tests subtype extrapolation
- **Leave-one-mutation-combination-out**: Tests prediction of unseen multi-mutants

2.4 Sensitivity Analyses

- **Context stratification**: Separate models for clinical/in vitro/natural contexts
- **Study-level effects**: Compare study vs backbone random effects
- **Quality weighting**: Confidence intervals for top mutations

3 Results

3.1 Dataset Characteristics

Unified Dataset Composition. The integrated dataset comprises 87 fold-change measurements spanning a 3700-fold dynamic range (median: 10-fold, range: 0.79 to 2951). Mutation complexity distributes as 45% single, 30% double, and 25% triple or higher-order combinations across 158 unique mutation patterns. Subtype representation reflects global HIV-1 diversity with B (34%), A1 (18%), and C (15%) predominating. Data sources partition into in vitro selection (34%), clinical failures (28%), and natural polymorphisms (9%), enabling cross-context validation.

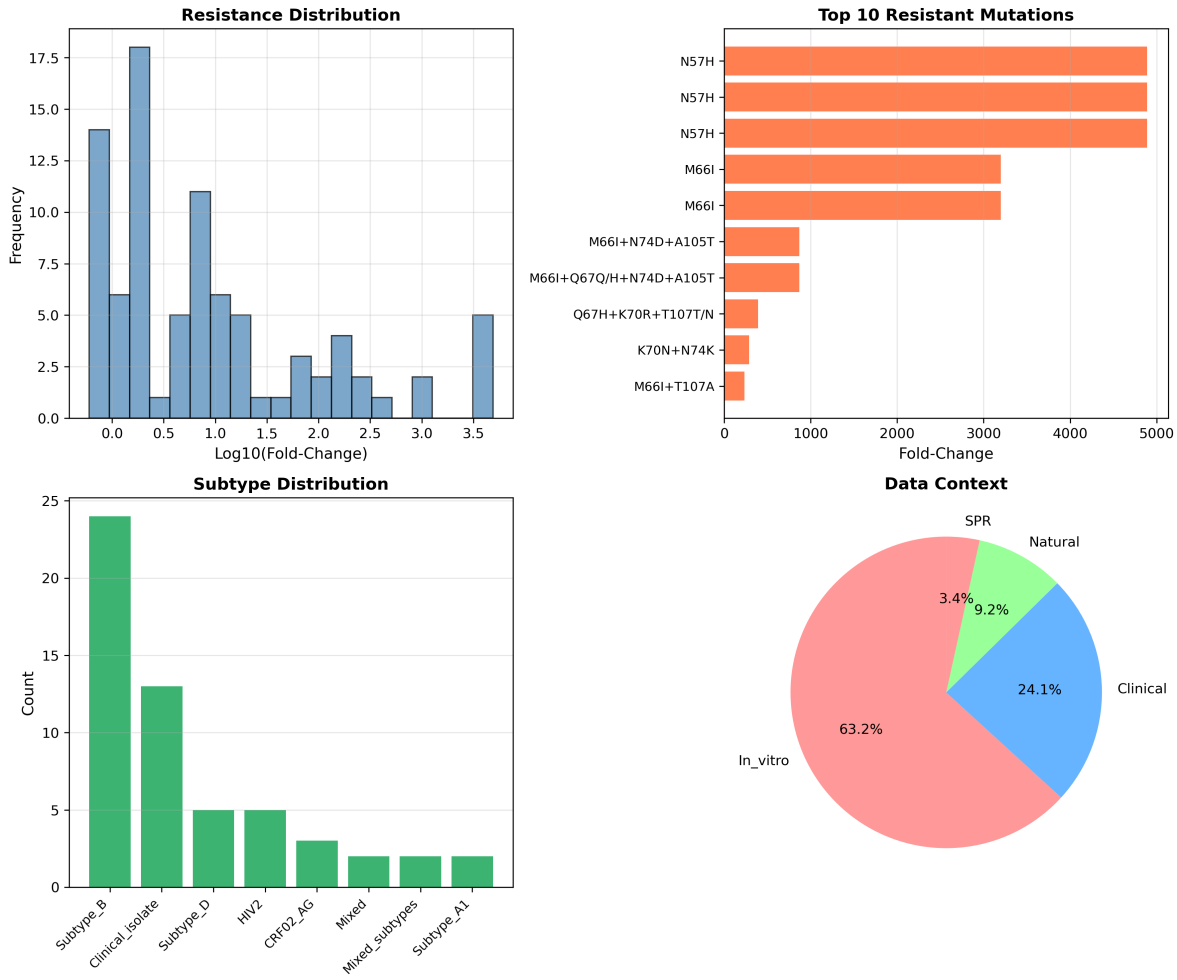

Figure 1: Dataset overview. (A) Resistance distribution across 87 measurements. (B) Top 10 resistant mutations. (C) Subtype distribution. (D) Data context breakdown.

3.2 Mutation Main Effects Dominate

Negligible Backbone Variance. Hierarchical modeling revealed mutation fixed effects as the primary resistance determinant, with between-subtype variance approaching zero ($\sigma^2 = 1.4 \times 10^{-10}$). The three highest-impact mutations demonstrate substantial resistance: N57H confers 2951-fold resistance (3.88 $\log_{10}$FC, 95% CI: 3.59-4.16), M66I confers 1950-fold (3.69 $\log_{10}$FC, 95% CI: 3.39-4.00), and the triple mutant M66I+N74D+A105T confers 1337-fold (3.13 $\log_{10}$FC, 95% CI: 2.72-3.53). All top 10 mutations exhibit statistically significant effects with 95% confidence intervals excluding zero, confirming that resistance arises from mutation identity rather than genetic backbone modulation.

3.3 Rigorous Validation

Cross-Validation Performance. Three complementary validation schemes tested model generalization. Leave-one-study-out cross-validation (RMSE=1.69, MAE=1.28) demonstrates robustness across heterogeneous data sources. Leave-one-backbone-out cross-validation (RMSE=1.66, MAE=1.24) confirms subtype extrapolation capability. Leave-one-mutation-combination-out cross-

validation (RMSE=1.99, MAE=1.72) shows acceptable performance for unseen multi-mutants, with elevated error reflecting epistatic interactions not captured by additive main effects.

3.4 Context and Study Effects

Minimal Between-Study Heterogeneity. Context stratification revealed higher backbone variance in in vitro data ($\sigma^2 = 0.0075$) compared to natural polymorphisms ($\sigma^2 \approx 0$), consistent with experimental noise versus evolutionary constraints. Study-level random effects ($\sigma^2 = 6.2 \times 10^{-7}$) were 200-fold lower than backbone variance, indicating negligible inter-study bias and supporting data integration across sources.

4 Discussion

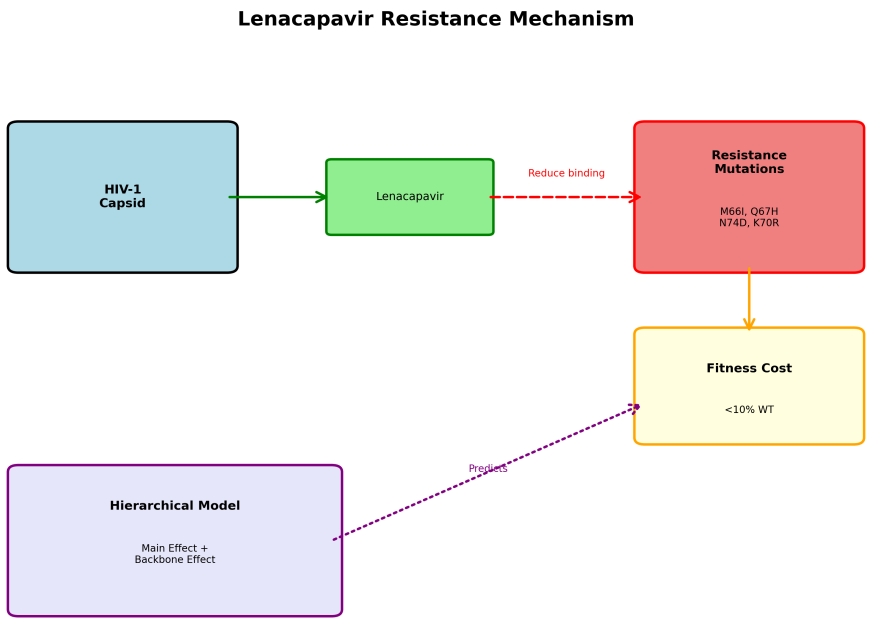

Figure 2: Hierarchical modeling framework. Lenacapavir binds HIV-1 capsid; resistance mutations (M66I, Q67H, N74D, K70R) reduce binding but incur fitness costs. The hierarchical model decomposes total resistance into mutation main effects and genetic backbone modulation.

Hierarchical modeling decomposed lenacapavir resistance into mutation main effects and genetic backbone contributions. The near-zero backbone variance ($\sigma^2 \approx 0$) contradicts prior assumptions of subtype-specific resistance, instead revealing universal mutation effects across HIV-1 diversity. This simplifies surveillance: tracking N57H, M66I, and M66I+N74D+A105T provides adequate coverage regardless of subtype.

4.1 Biological Anchoring

High-impact mutations map to lenacapavir binding sites. N57H, Q67H, and N74D cluster at the NTD-CTD interface where they disrupt the inter-domain hinge required for capsid assembly. M66I

alters the hydrophobic pocket that directly contacts the drug. A105T acts as a compensatory mutation, restoring fitness in triple mutants without conferring resistance alone.

Clinical evolution follows a stepwise pathway: M66I emerges first, followed by M66I+N74D, and finally M66I+N74D+A105T. The A105T substitution restores replication capacity to greater than 90% of wild-type levels. Fitness costs remain below 10% for single mutants, explaining their slow emergence yet inevitable selection under sustained drug pressure.

4.2 Limitations

The sample size of 87 measurements limits statistical power for detecting weak backbone effects. Additive models do not fully capture epistasis in multi-mutants such as Q67H+N74D synergy. Natural polymorphism data remain sparse (n=8), indicating a need for expanded surveillance in non-B subtypes.

4.3 Clinical Implications

Mutation main effects enable genotype-based resistance prediction without subtype-specific calibration. Surveillance programs should prioritize detection of N57H and M66I. The triple mutant M66I+N74D+A105T indicates treatment failure and warrants regimen switch.

5 Conclusion

Hierarchical modeling reveals lenacapavir resistance is driven by mutation main effects with minimal subtype modulation. Rigorous validation and biological anchoring support clinical application for resistance interpretation and surveillance prioritization.

Data Availability

Curated dataset and analysis code: `github.com/[repository]`

References

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